

Kunal Maheshwari

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SUMMARY

A seasoned Software Engineer with over 4 years of experience specializing in scalable Java/Spring backend systems and cloud-native data pipelines. Expertise spans healthcare, automotive, and research domains with a strong command of Java 8/11, Spring Boot, and distributed systems. Proven ability to engineer hybrid-cloud integrations using Azure and AWS, and deploy containerized data workflows with Spark and Docker. Certified Azure engineer with a focus on big data and privacy-preserving ML systems, adept at Agile methodologies and team mentorship, providing strong technical support and solution design.

EDUCATION

Central Michigan University | Mount Pleasant, MI, USA Aug 2024 – Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 – May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 – August 2024

- Designed and developed SOA-based backend services using Java 8/11 and SOAP APIs for healthcare insurance, deployed on IBM WebSphere Application Server, maintaining 99.99 percent SLA availability for 50,000+ daily API requests.
- Integrated REST and SOAP services through an API Gateway using certificate-based authentication, validating requests and responses for downstream service orchestration.
- Engineered an Azure Relay-based integration utilizing Azure Functions, publishing messages to Azure Service Bus topics and forwarding them to JMS queues, demonstrating robust cloud-native solution design.
- Led the modernization of logging infrastructure by migrating to Log4j 2.x, establishing standardized configurations and upgrading multiple production services, enhancing overall system visibility.
- Collaborated within Agile teams to perform SOAP and REST service validation using SOAP UI, verify data correctness through database inspection, and troubleshoot production issues, providing critical technical support.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 – August 2022

- Developed enterprise Java and EJB applications for automotive clients, including RESTful services and monolithic systems built using Server Faces (JSF).
- Engineered vehicle configuration modules that processed option codes via REST calls to a Python-based decision tree model, ensuring business rule compliance.
- Optimized SQL queries and refactored legacy Java code within high-traffic systems, improving overall system performance by approximately 90 percent.
- Implemented and maintained CI/CD pipelines using Jenkins and GitLab, automating builds, testing, and quality validation for streamlined software engineering processes.
- Mentored junior developers through code reviews and debugging sessions, emphasizing clean coding and modular design principles.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 – October 2021

- Maintained high-availability automotive systems, ensuring over 99 percent uptime through proactive monitoring and backend performance optimization.
- Enhanced SQL-driven data pipelines and backend operations, improving query execution efficiency and strengthening transactional data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving high code coverage and improving release stability.
- Supported sprint planning, release coordination, and defect resolution within Agile teams using JIRA-based workflows.

Trainee | Capgemini | Jalandhar, Punjab, India January 2020 – July 2020

- Contributed to backend development of a Fitness Club web application, implementing user profile management and secure email-based authentication workflows.
- Developed CRUD-based transactional features for an Online Wallet application, supporting account management, transaction tracking, and balance updates.
- Built backend services using Java Spring Framework with Angular frontend integration and PostgreSQL database support.
- Participated in code reviews and feature testing within iterative development cycles to deliver functional modules.

PROJECTS

NYU Taxi Data Analysis Platform August 2024 – December 2025

- Architected a cloud-native, distributed Spark data pipeline to process and analyze over 8 GB of NYC taxi trip data using PySpark.
- Implemented an event-driven ingestion and processing workflow using AWS Lambda and EventBridge to trigger containerized Spark jobs on ECS, demonstrating strong solution design.
- Deployed Spark workloads on ECS with shared storage via EFS, enabling scalable execution and efficient handling of large datasets within Linux environments.
- Automated end-to-end cloud infrastructure provisioning using Terraform, including ECS clusters, task definitions, IAM roles, and storage resources.

Privacy-Preserving AI Email Client (Capstone Project)

- Engineered an end-to-end privacy-preserving email intelligence system using finetuned T5 sequential transformer models for thread-aware summarization and classification, exposed via production-ready RESTful APIs.

- Designed a per-user fine-tuning pipeline using parameter-efficient LoRA adapters, enabling isolated model training and inference without sharing data or model state across users.
- Implemented secure local deployment with model quantization and orchestration, dynamic model loading, and LRU-based memory management to support low-resource environments.
- Built a backend orchestration layer to manage model lifecycle, caching, and inference concurrency, improving reliability and runtime efficiency. Provided critical technical support for model deployment and maintenance.

SKILLS

- Programming Languages: Proficient in Java (8/11), Python, and SQL, with experience in Linux/Unix environments.
- Frameworks and Libraries: Expertise in Spring Boot, Spring Framework, RESTful API, SOAP API, SOA, microservices, JPA, Hibernate, Json, Yaml, OpenAPI, Postman, XML, JDBC, Apache Spark, PySpark, JUnit, Log4j 2.x, Server Faces (JSF), J2EE, EJB, and Windows-based development.
- Cloud Platforms: Certified Azure engineer with extensive experience in AWS (EC2, RDS, S3, Lambda, ECS, EFS, EventBridge), Azure (Functions, DevOps, ServiceBus, Relay), and GCP (Pub Sub), focusing on solution design and technical support.
- Databases and DevOps: Skilled in PostgreSQL, Oracle Database, IBM DB2, IBM WebSphere, SQL, Docker, Kubernetes, Terraform, Jenkins, Git, SVN, Maven, JIRA, Confluence, IBM UrbanCode Deploy, and DBeaver, emphasizing robust software engineering practices.
- Data Analysis and Methodologies: Experienced in Apache Spark, PySpark, ETL Pipelines, Data Warehousing, Data Analytics, Tableau, Big Data Analytics, Agile Development, System Design, Unit Testing, Scrum, Test-Driven Development, Object Oriented Programming, and CI/CD, ensuring effective communications and solution design.